Customer Segmentation & Targeting Guide

A comprehensive guide to the Customer Segmentation module for OpenCart 3.0.3.8. This module enables advanced customer group management, personalized frontend experiences, and automated push notifications through Firebase integration.

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Quick Start: Installation Options

Depending on your needs, you can either run the full pre-configured sample project or install the plugin standalone on an existing OpenCart site.

Option A: Full Sample Project (Recommended for Testing)

Use the **`opencart_customer_segmentation_sample.zip`** package. This contains a complete, containerized OpenCart environment.

1. **Extract** the ZIP file onto your machine.
2. **Launch**: Open your terminal in the project directory and run:

   ```
   docker-compose up --build
   ```

3. **Access**:
   - **Store Frontend**: `http://localhost/`
   - **Admin Panel**: `http://localhost/admin/` (Login: `opencart` / `opencart`)
   - **Database (phpMyAdmin)**: `http://localhost:8080/`

Option B: Standalone Plugin Installation

Use the **`customer_segmentation.ocmod.zip`** package if you already have an existing OpenCart 3.0.3.8 site.

1. **Extract** the plugin ZIP locally.
2. **Upload**: Copy the contents of the `upload/` folder to your OpenCart root directory (preserving the directory structure).
3. **Installer Option**: Alternatively, go to **Extensions -> Installer** and upload the `customer_segmentation.ocmod.zip` file directly.

4. **Install**:
    - Go to **Extensions -> Extensions -> Modules**.
    - Find **Segmentation & Targeting (Manager)** and click **Install**.
    - Click **Edit** to configure settings and rebuild the database schema (automatic on first run).
5. **Events**: The module automatically registers the necessary event hooks upon installation.

Overview

The Customer Segmentation module is a powerful tool designed to help store owners deliver targeted content and automated marketing actions based on customer behavior and attributes.

Customer Group Base

> **Important**
>
> The segmentation feature is built directly on top of OpenCart's internal **Customer Groups**. All rules eventually result in moving a customer between these groups, which then controls the visibility of personalized banners, sliders, and promotions.

Key Features

- **Dynamic Segmentation**: Automatically reassign customers to different groups based on custom rules (e.g., total spend, order frequency).
- **Personalized UI**: Display specific banners, product sliders, and promotional boxes to different customer segments.
- **Push Notifications**: Integrated Firebase support for sending automated push notifications when customers enter new segments.
- **Seamless Management**: Integrated logs and a "Mock Notifications App" for easy testing.

Relevant Links

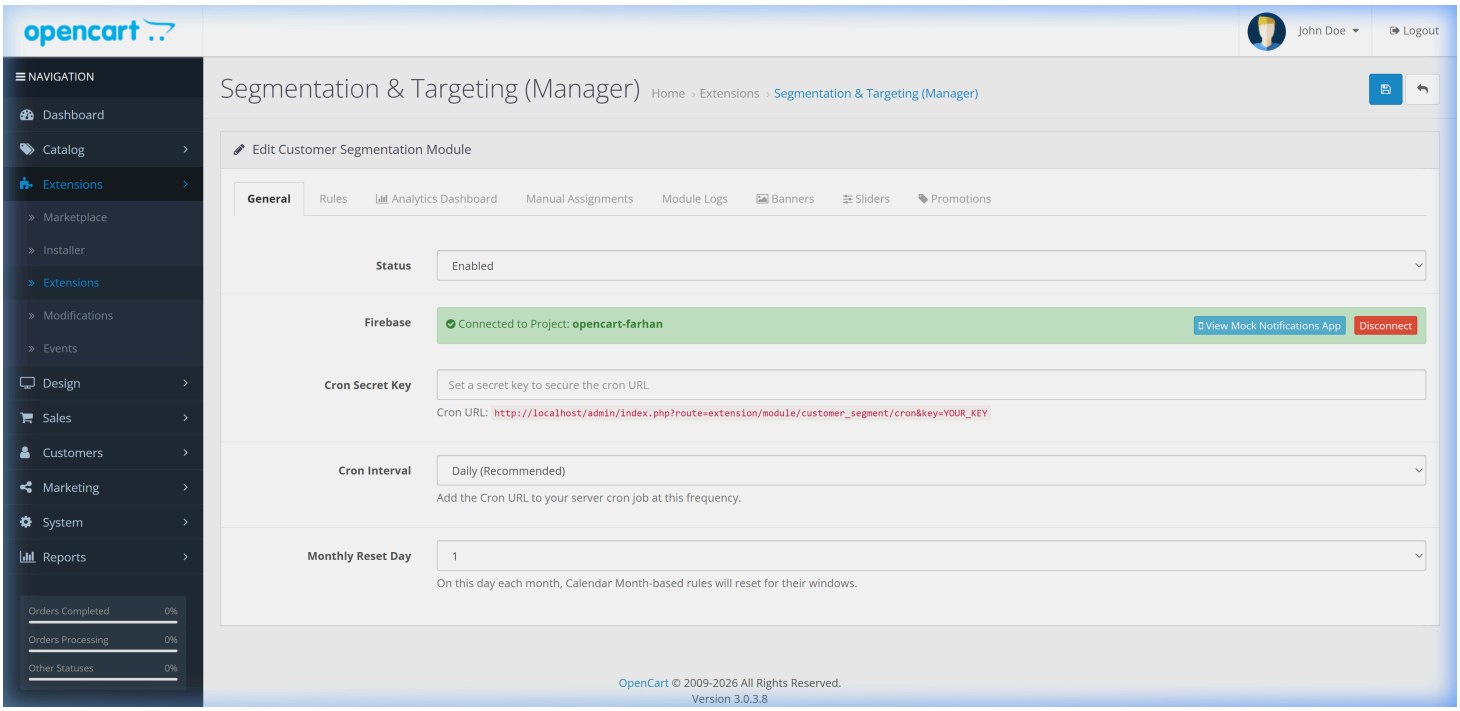
- Firebase Console
- OpenCart User Manual
- Firebase Cloud Messaging Documentation

Administrative Configuration

The module's main management interface is located under **Extensions -> Extensions -> Modules -> Segmentation & Targeting (Manager)**.

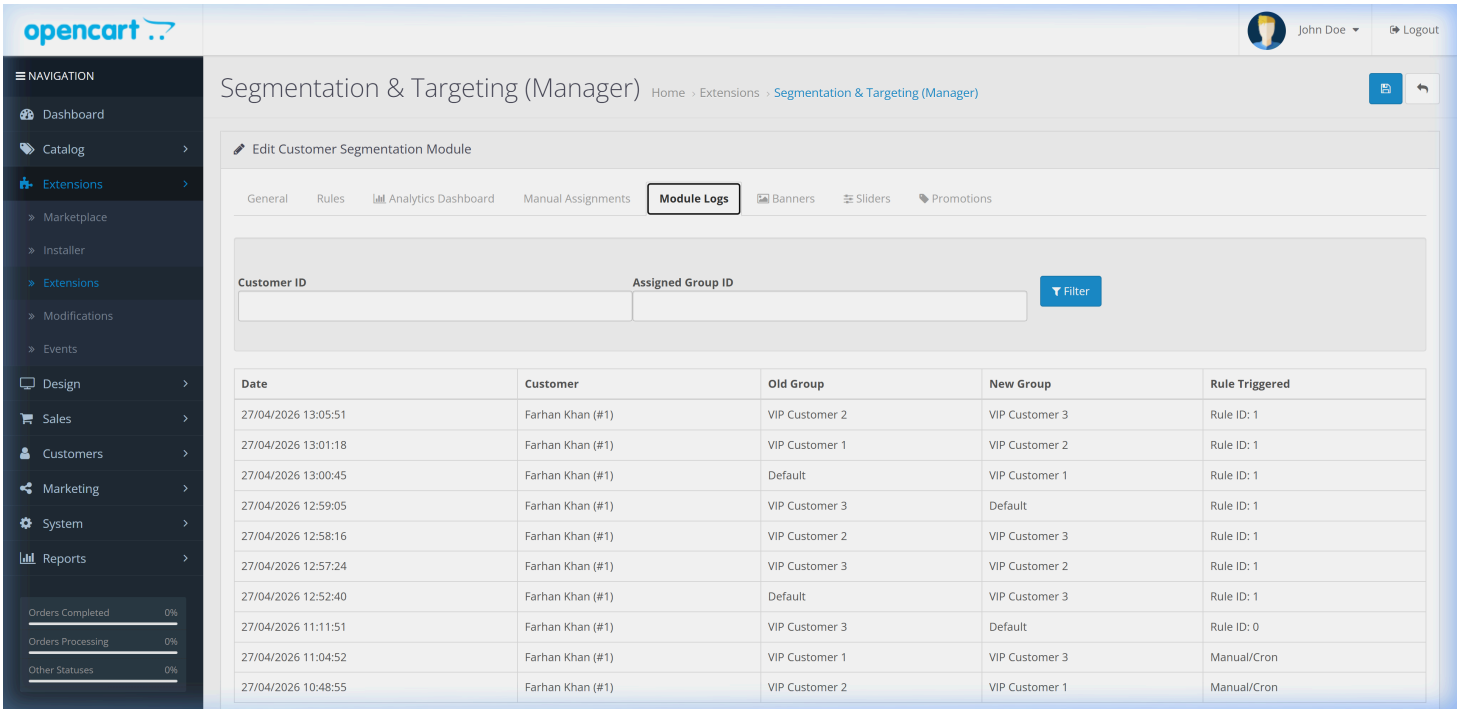
General Settings

In the **General** tab, you can enable or disable the module and configure the core Firebase connection.



Connectivity & Logs

- **Firebase Connection**: Upload your Firebase Service Account JSON to establish a secure connection.
- **Module Logs**: Monitor all segmentation activities, group reassignments, and notification triggers in the **Logs** tab.

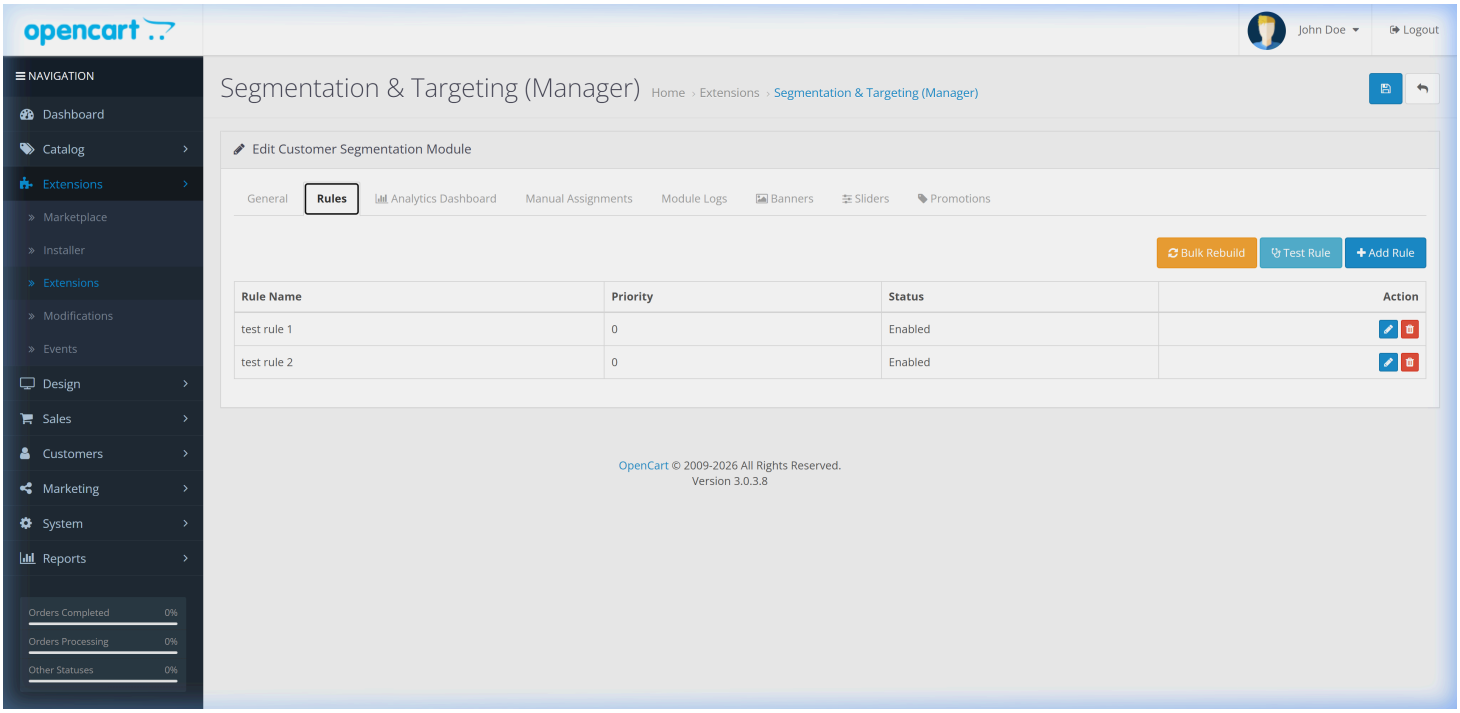


Rule Engine

The heart of the module is the Rule Engine, which allows you to define how and when customers are segmented.

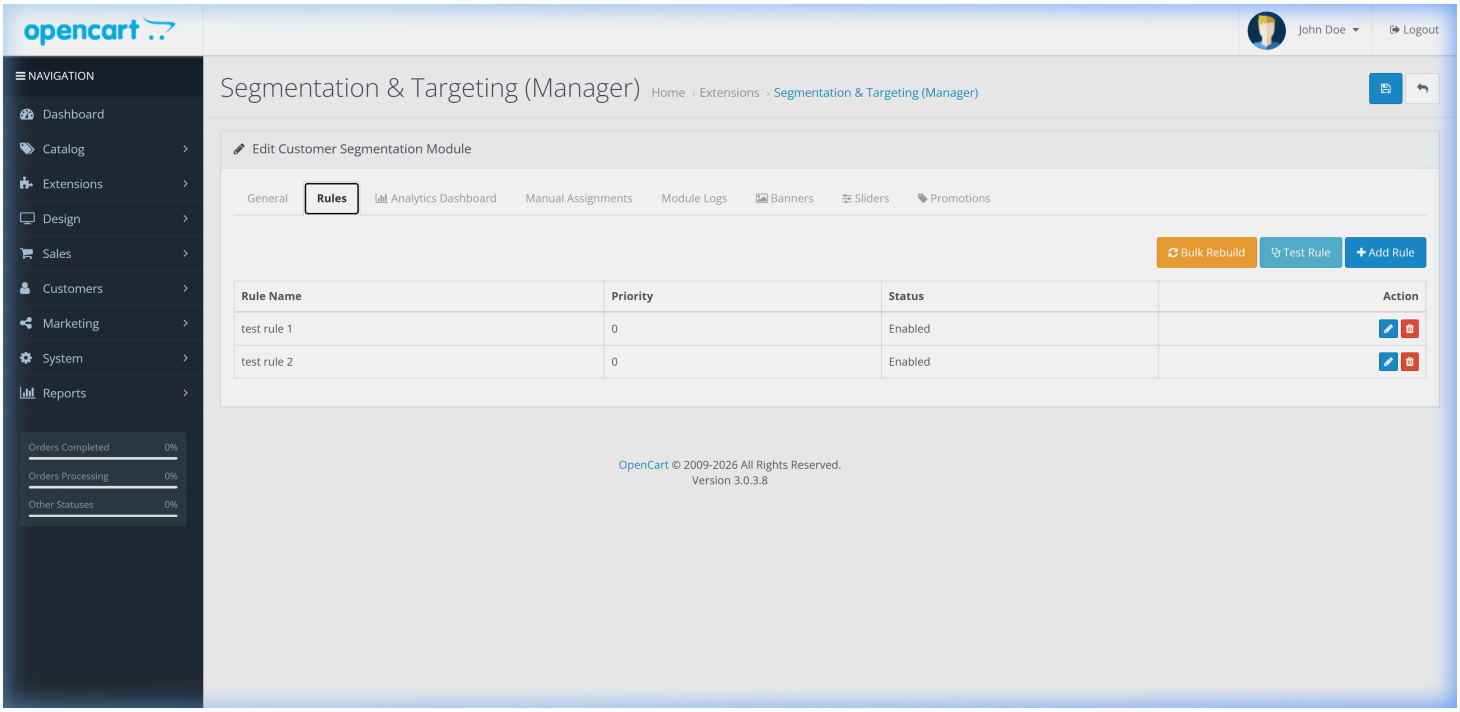
Managing Rules

Rules are listed in the **Rules** tab. You can set priorities to determine which rule takes precedence if multiple rules match a customer.



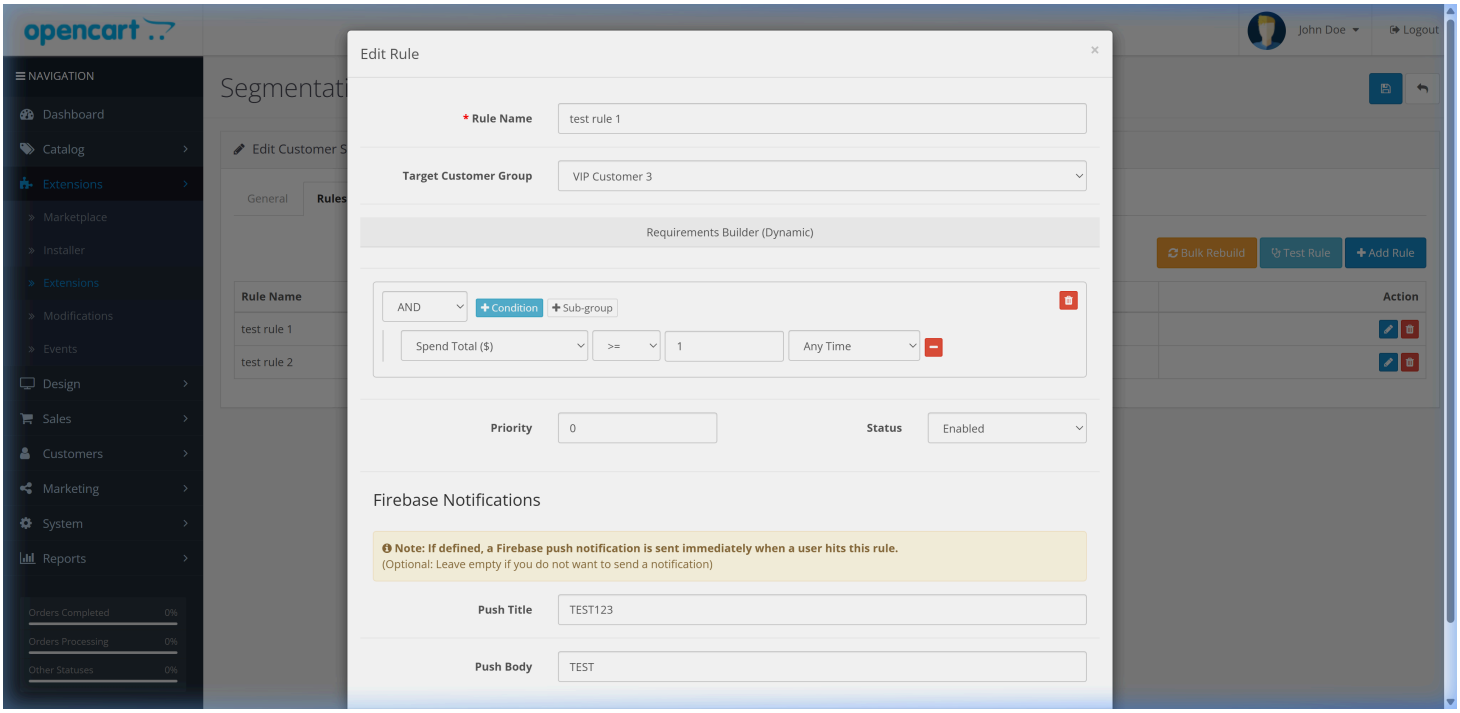
Bulk Rebuild Feature

The **Bulk Rebuild** button allows you to re-evaluate all existing customers against all active rules simultaneously. This is essential after creating new rules or modifying existing ones to ensure the entire database is correctly segmented.



Rule Configuration

Each rule consists of **Triggers**, **Conditions**, and **Actions**.



- **Triggers**: Define the event that starts the evaluation (e.g., "After Checkout", "Daily Cron").
- **Conditions**: Use the "Requirements Builder" to set specific criteria (e.g., "Total Orders > 5", "Registered within last 30 days").
- **Actions**: Define what happens when a match is found (e.g., "Reassign Group", "Send Notification").

Firebase Integration & Push Notifications

The module uses Firebase Cloud Messaging (FCM) for real-time engagement.

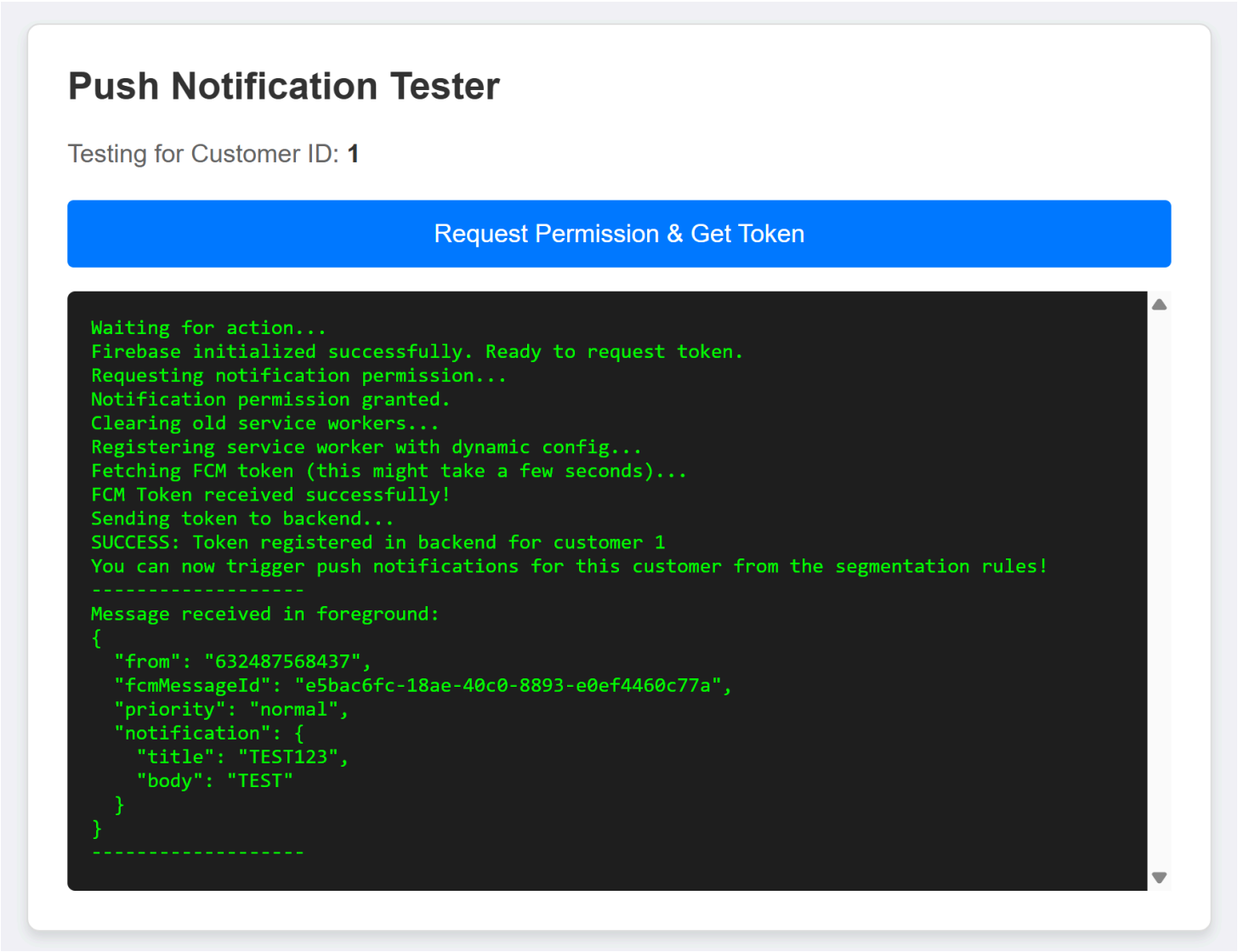
Setup Process

1. Create a project in the Firebase Console.
2. Generate a **Service Account JSON** file and upload it in the module's General settings.
3. Configure your **VAPID Key** for web push notifications.

Mock Notifications App (Testing)

To test your setup without a live frontend app, use the integrated **Mock Notifications App**.

1. Navigate to the rule you want to test.
2. Open the testing modal and click "Open Mock App".
3. Request notification permission to register a test token.
4. Trigger the rule action to see the notification arrive instantly.



Personalized Frontend Widgets

Deliver the right message to the right person using specialized layout modules.

Component Types

1. **Banners**: Display targeted promotions or welcome messages.
2. **Product Sliders**: Show recommended products based on the customer's segment.
3. **Promotions**: Special offers with descriptions and optional coupon highlights.

Configuration

Manage targeted content within the Manager module, then assign the corresponding layout modules in **Design -> Layouts**.

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John Doe | Logout

NAVIGATION

- Dashboard
- Catalog
- Extensions
  - Marketplace
  - Installer
  - Extensions
  - Modifications
  - Events
- Design
- Sales
- Customers
- Marketing
- System
- Reports

Orders Completed 0%
Orders Processing 0%
Other Statuses 0%

Segmentation & Targeting (Manager)

Home › Extensions › Segmentation & Targeting (Manager)

Edit Customer Segmentation Module

General | Rules | Analytics Dashboard | Manual Assignments | Module Logs | **Banners** | Sliders | Promotions

Filter by Group

All Groups

Filter

Add Banner

ID	Group	Title	Image	Link	Status	Action
2	4	------ USER GROUP 3 ------			Enabled	

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Edit Customer Segmentation Module

General | Rules | Analytics Dashboard | Manual Assignments | Module Logs | Banners | **Sliders** | Promotions

Filter by Group

All Groups

Filter

Add Slider

ID	Group	Name	Product IDs	Status	Action
1	4	We think you will like these	30,47,41,42,28	Enabled	

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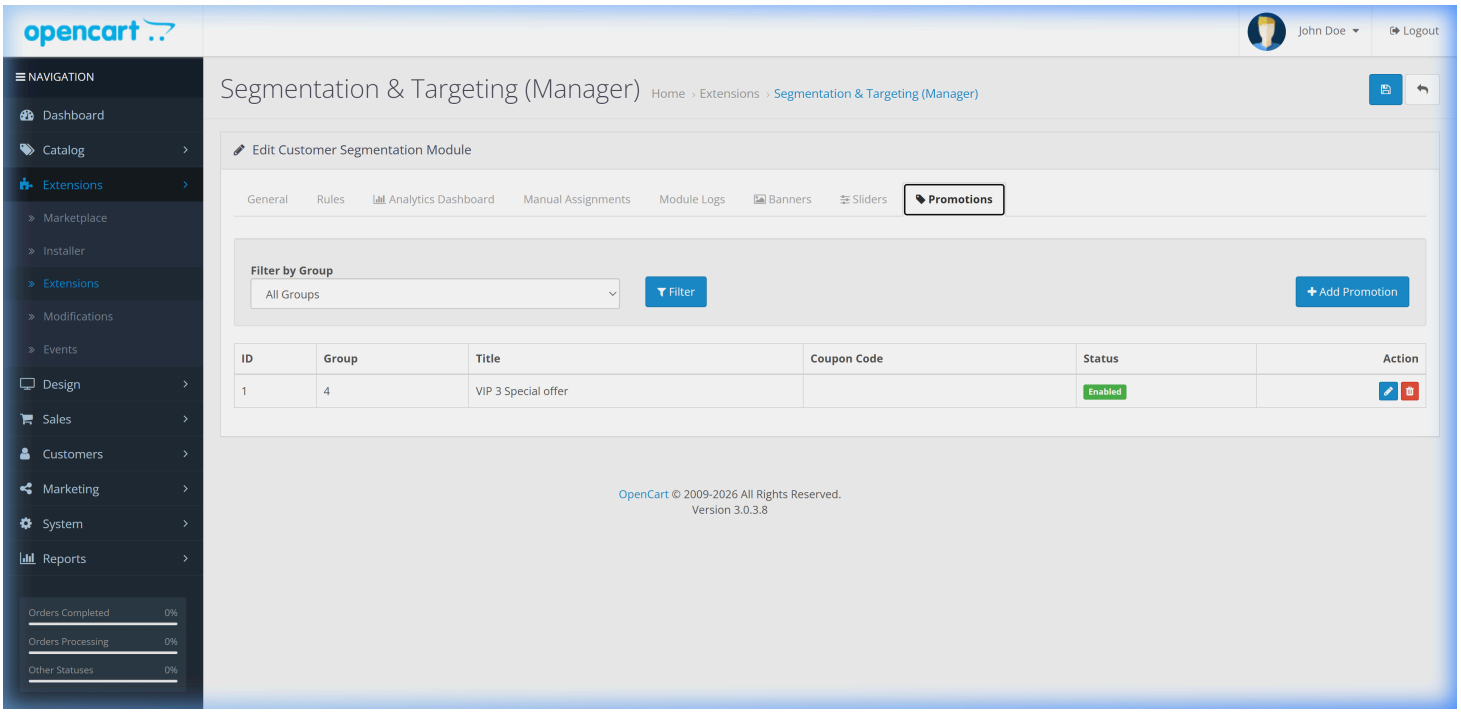

Frontend & Mobile APIs

The module provides specialized JSON endpoints categorized into customer-facing (Catalog) and management-facing (Administrative) APIs.

Catalog APIs (Customer-Facing)

These endpoints are used by the storefront or mobile app. They require a valid customer session.

1. Get Personalized Content

Returns segments-specific content for the logged-in customer.

- **URL**: `index.php?route=extension/module/customer_segment/getPersonalized`
- **Method**: `GET`
- **Response**: JSON object containing `banners`, `sliders`, and `promotions`.

2. Save FCM Token

Registers a Firebase token for push notifications.

- **URL**: `index.php?route=extension/module/customer_segment/saveToken`
- **Method**: `POST` (Param: `token`)

3. Trigger Calculation (Cron)

Securely triggers the rule evaluation engine.

- **URL**: `index.php?route=extension/module/customer_segment/cron&key=YOUR_KEY`
- **Method**: `GET`

Administrative APIs (Management)

These endpoints are used for building custom administrative dashboards or integrations. They require a valid `user_token` (Admin Session).

Rule Management

- **List Rules**:
  `GET admin/index.php?route=extension/module/customer_segment/getRules&user_token=...`
- **Get Rule Details**:
  `GET admin/index.php?route=extension/module/customer_segment/getRule&rule_id=X&user_token=...`
- **Save Rule**:
  `POST admin/index.php?route=extension/module/customer_segment/saveRule&user_token=...`
- **Delete Rule**:
  `GET admin/index.php?route=extension/module/customer_segment/deleteRule&rule_id=X&user_token=...`
- **Test Rule**:
  `POST admin/index.php?route=extension/module/customer_segment/testRule&user_token=...` (Params: `rule_id`, `customer_id`)

Segment & content Management

- **Manual Assignments**: `GET .../getManuals`, `POST .../addManual`, `GET .../deleteManual`
- **Banners**: `GET .../getBanners`, `POST .../saveBanner`, `GET .../deleteBanner` (Supports filtering by `customer_group_id`)
- **Sliders**: `GET .../getSliders`, `POST .../saveSlider`, `GET .../deleteSlider`
- **Promotions**: `GET .../getPromotions`, `POST .../savePromotion`, `GET .../deletePromotion`

Analytics & Utilities

- **Activity Logs**: `GET .../getDynamicLogs` (Supports pagination and filtering)
- **Bulk Rebuild**: `GET .../bulkRebuild&start=0` (Processes in batches of 50)
- **Firebase Connectivity**: `POST .../verifyFirebase`, `POST .../disconnectFirebase`
- **Helpers**: `GET .../autocomplete` (Customer search), `GET .../getProductInfo` (Multi-product details)

Developer Information

Database Schema

The module introduces several new tables to support its functionality:

Table Name	Description
`customer_segment_rule`	Stores rule definitions, priorities, and conditions.
`customer_segment_fcm_token`	Stores device tokens for push notifications.
`customer_segment_log`	Audit trail for group changes and rule triggers.
`customer_segment_manual`	Stores manual segment overrides and locks.
`customer_segment_banner`	Stores targeted banner content linked to groups.
`customer_segment_slider`	Stores targeted product slider configurations.
`customer_segment_promotion`	Stores targeted promotional messages and codes.

Additionally, the core `customer` table is extended with:

- `date_of_birth` : Supports age-based segmentation.
- `gender` : Supports demographic targeting.

Event Hooks

The module utilizes OpenCart's event system to trigger real-time evaluations:

- `catalog/model/checkout/order/addOrderHistory/after` : Evaluates rules after a successful checkout.
- `catalog/controller/common/footer/after` : Injects the service worker initialization for push notifications.